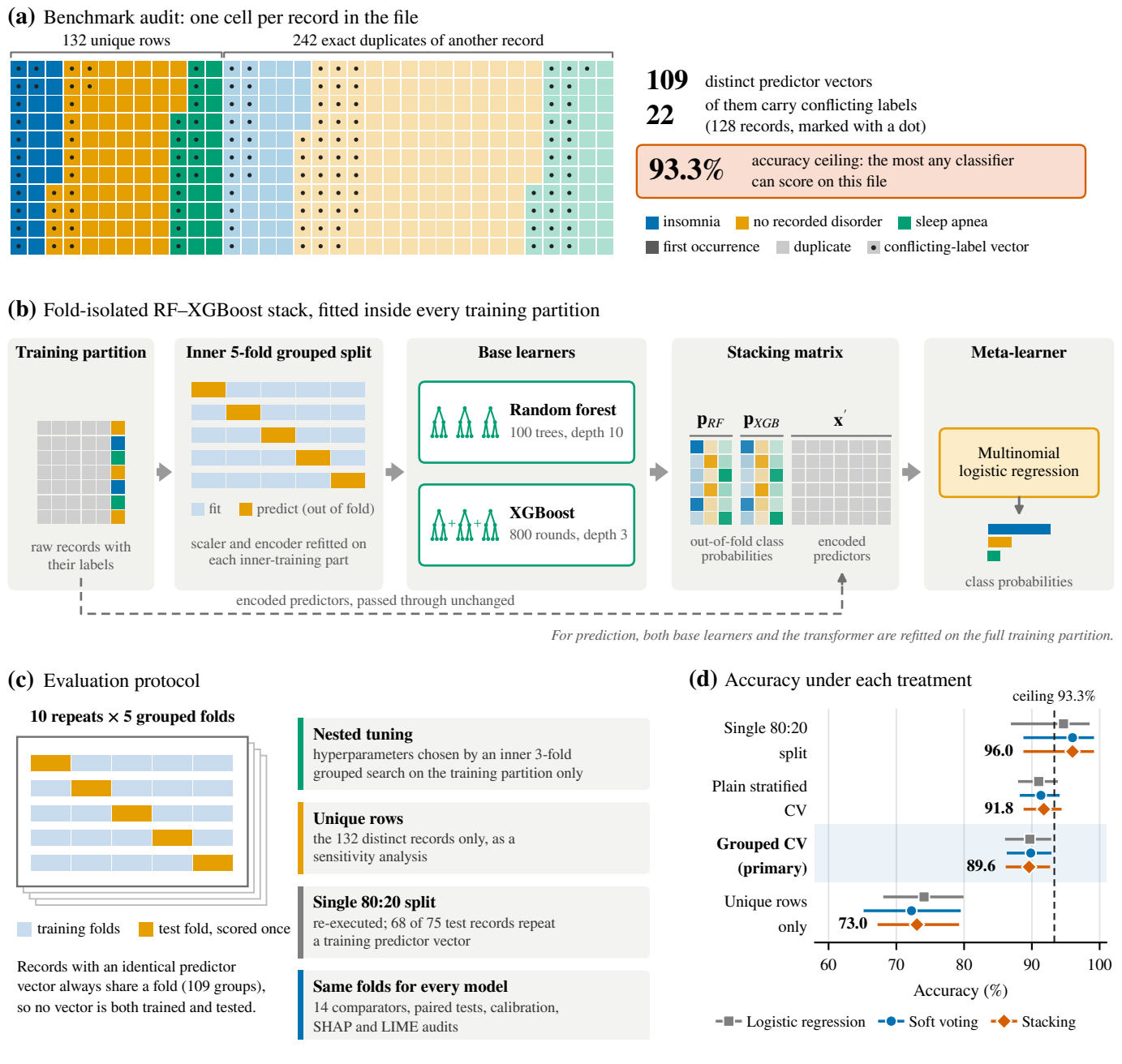

(a) Benchmark audit: one cell per record in the file
132 unique rows
242 exact duplicates of another record
109 distinct predictor vectors
22 of them carry conflicting labels
(128 records, marked with a dot)
93.3% accuracy ceiling: the most any classifier can score on this file
insomnia
no recorded disorder
sleep apnea
first occurrence
duplicate
conflicting-label vector
(b) Fold-isolated RF–XGBoost stack, fitted inside every training partition
Training partition
raw records with their labels
Inner 5-fold grouped split
fit
predict (out of fold)
scaler and encoder refitted on each inner-training part
Base learners
Random forest
100 trees, depth 10
XGBoost
800 rounds, depth 3
Stacking matrix
p_{RF}
p_{XGB}
x'
out-of-fold class probabilities
encoded predictors
Meta-learner
Multinomial logistic regression
class probabilities
encoded predictors, passed through unchanged
For prediction, both base learners and the transformer are refitted on the full training partition.
(c) Evaluation protocol
10 repeats × 5 grouped folds
training folds
test fold, scored once
Records with an identical predictor vector always share a fold (109 groups), so no vector is both trained and tested.
Nested tuning
hyperparameters chosen by an inner 3-fold grouped search on the training partition only
Unique rows
the 132 distinct records only, as a sensitivity analysis
Single 80:20 split
re-executed; 68 of 75 test records repeat a training predictor vector
Same folds for every model
14 comparators, paired tests, calibration, SHAP and LIME audits
(d) Accuracy under each treatment
ceiling 93.3%
Single 80:20 split
96.0
Plain stratified CV
91.8
Grouped CV (primary)
89.6
Unique rows only
73.0
60
70
80
90
100
Accuracy (%)
Logistic regression
Soft voting
Stacking